

New [blog post](#) on the May 13–15 incident. We sincerely apologize for the incident, the disruption it caused, and any concern it raised.

There is a [newer version](#) of the record available.

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Relaxation-Driven Cyclic Cosmology (RDCC) v21.0 - solves the near-adiabaticity puzzle

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This document presents v21.0 of Relaxation-Driven Cyclic Cosmology (RDCC),

a minimal and nonsingular cosmological framework where the Universe undergoes an infinite sequence of smooth contraction and expansion phases. Formulated entirely within standard General Relativity and a single complex scalar field, RDCC incorporates the leading Planck-suppressed interaction between two CPT-conjugate sectors sharing the same spacetime metric. This two-sector structure replaces the assumption of a globally directed time with a CPT-symmetric description, making thermodynamic arrows, entropy production, and matter-antimatter asymmetries emergent, sector-dependent phenomena. The global system remains entropy-conserving, resolving the Tolman entropy problem without inflation, modified gravity, or additional fields.

The scalar potential features a radiatively tilted double-well structure augmented by an ekpyrotic term dominant during contraction. Combined with bounded effective Friedmann corrections from the EFT expansion, these elements produce a smooth, ghost-free bounce with finite curvature invariants. The ekpyrotic phase generates a nearly scale-invariant spectrum of curvature perturbations, while the Planck-suppressed cross-sector interaction damps isocurvature modes as $a^{-\alpha}$, driving the Universe toward near-adiabaticity.

A key outcome of the relaxation mechanism is a temperature asymmetry between sectors, naturally yielding the observed dark-matter-to-baryon ratio through exponential sensitivity to the relaxation parameter α . This

parameter also induces small, correlated deviations from Λ CDM in growth rates and the matter power spectrum, plus a modest ΔN_{eff} contribution. RDCC provides a unified explanation for nonsingular evolution, dark matter, baryon asymmetry, the thermodynamic arrow, and primordial near-adiabaticity. Its predictions, derived from a single infrared parameter, are testable with upcoming large-scale structure and CMB experiments, offering a structurally simple yet observationally rich alternative to standard cosmology.

Changes from v20.0: Enhanced emphasis on EFT consistency with dimension-8 operators and positivity bounds; explicit resolution of the near-adiabaticity puzzle via isocurvature damping; refined perturbation theory and phenomenological predictions.

This version builds on:

Relaxation-Driven Cyclic Cosmology (RDCC) v1.0: A Minimal Eternal Framework and Its Dynamical Realization: <https://zenodo.org/records/18281686>

Relaxation-Driven Cyclic Cosmology (RDCC) v2.0: A Minimal Eternal Framework with Dynamical Realization and Extensions: <https://zenodo.org/records/18304744>

Relaxation-Driven Cyclic Cosmology (RDCC) v5.0: Bounce-Matching Formalism, Mirror-Sector Dynamics, and Refined Gravitational Waves: <https://zenodo.org/records/18307836>

Relaxation-Driven Cyclic Cosmology (RDCC) v6.0: Black Hole Bounces as Local CPT-Fixpoints, Full Perturbation Transfer Matrix and Mirror-Sector Microphysics: <https://zenodo.org/records/18312656>

Relaxation-Driven Cyclic Cosmology (RDCC) v7.0: Quantum-Corrected Black Hole Bounces, Full Perturbation Transfer Matrix with Phase-Shift, and Mirror Microphysics <https://zenodo.org/records/18327084>

Relaxation-Driven Cyclic Cosmology (RDCC) v8.0: Parameter Space Exploration, Data Confrontation, and Predictive Structure Formation <https://zenodo.org/records/18334398>

Relaxation-Driven Cyclic Cosmology (RDCC) v9.0: Predictive Cosmology, Nonlinear Structure Formation, and Astrophysical Signatures <https://zenodo.org/records/18336237>

Relaxation-Driven Cyclic Cosmology (RDCC) v10.0: Unified Cosmology, Full Simulations, and Observational Pipelines <https://zenodo.org/records/18336865>

Relaxation-Driven Cyclic Cosmology (RDCC) v11.0: Entropy Dynamics, Mirror Thermodynamics, EFT Consistency, and Naturalness <https://zenodo.org/records/18341147>

Relaxation-Driven Cyclic Cosmology (RDCC) v12.0: Precision Predictions, Multi-Probe Confrontation, and Quantum-Geometric Completion <https://zenodo.org/records/18346555>

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Relaxation-Driven Cyclic Cosmology (RDCC) v15.0: Mirror-Sector Cosmology, Hidden-Universe Gravitational Waves, and Two-Sector Evolution <https://zenodo.org/records/18358095>

Relaxation-Driven Cyclic Cosmology (RDCC) v16.0: Precision Two-Sector Dynamics, Mirror Thermodynamics Refinements, and Enhanced GW Forecasts <https://zenodo.org/records/18388123>

Relaxation-Driven Cyclic Cosmology (RDCC) v17.0 - Minimal Effective Two-Sector Version
<https://zenodo.org/records/18761806>

While the core of RDCC remains a structural, timeless framework with fixpoint character, this document explores a dynamical realization of the bounce mechanism to enable concrete, falsifiable predictions.

Related documents and the full RDCC ecosystem are available via Zenodo:
<https://zenodo.org/records/18204087>

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Relaxation-Driven Cyclic Cosmology (RDCC) v21.0 -
Minimal Effective Two-Sector Version

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) is a minimal and nonsingular cosmological framework in which the Universe undergoes an infinite sequence of smooth contraction and expansion phases. The theory is formulated entirely within standard General Relativity and a single complex scalar field, supplemented only by the leading Planck-suppressed interaction between two CPT-related sectors that share the same spacetime metric. This two-sector structure replaces the assumption of a globally directed time with a CPT-symmetric description in which thermodynamic arrows, entropy production and matter-antimatter asymmetries emerge as sector-dependent phenomena. The global system remains entropy-conserving, thereby resolving the Tolman entropy problem without invoking inflation, modified gravity or additional fields.

The scalar potential consists of a radiatively tilted double-well structure and an ekpyrotic term that dominates during contraction. Together with the bounded effective Friedmann correction implied by the EFT expansion, these ingredients generate a smooth, ghost-free bounce with finite curvature invariants. The ekpyrotic phase produces a nearly scale-invariant spectrum of curvature perturbations, while the Planck-suppressed cross-sector interaction damps isocurvature modes as a^{-6} , driving the Universe toward adiabaticity. The resulting spectral index and amplitude are consistent with current CMB observations.

A key phenomenological consequence of the relaxation mechanism is the emergence of a temperature asymmetry between the two sectors, which naturally generates the observed dark-matter-to-baryon ratio through an exponential sensitivity to the relaxation parameter α . The same parameter controls small, correlated deviations from Λ CDM in the growth rate and matter power spectrum, as well as a modest contribution to the effective number of relativistic species. These signatures provide a clear and falsifiable observational window into the two-sector structure.

RDCC thus offers a unified and minimal explanation for nonsingular cosmological evolution, dark matter, the baryon asymmetry, the thermodynamic arrow and the near-adiabaticity of primordial perturbations. Its predictions arise from a single infrared parameter and are testable with upcoming large-scale-structure and CMB experiments, making RDCC a structurally simple yet observationally rich alternative to standard cosmology.

1 Introduction

The standard Λ CDM model provides an excellent description of the expansion history and the statistical properties of cosmological perturbations. Yet several foundational questions remain unresolved: the nature of dark matter and dark energy, the origin of the baryon asymmetry, the thermodynamic arrow of time, the near-adiabaticity of primordial perturbations and the presence of an initial singularity. These puzzles are typically addressed by introducing new degrees of freedom, modified gravity or an inflationary phase. While such extensions can resolve individual issues, they do not explain why these problems appear together or why their proposed solutions remain conceptually disconnected.

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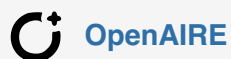
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About

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Infrastructure

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Projects

Roadmap

Contact

Blog

Blog

Support

Help

FAQ

Developers

REST API

OAI-PMH

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